

A non-superreflexive example with Frechet smooth canonical ultrapower points

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Status. `partial_result_likely_valid`. The result below is a partial answer to the smoothness question raised in arXiv:1412.1279: it shows that the hypothesis does not force superreflexivity.

Source question. Talponen asks what geometric properties are implied by the Frechet smoothness of the canonical ultrapower point $j(x)$, for every $x \in S_X$ [1, p. 4].

all Gâteaux smooth ultrapowers are uniformly smooth.

□

As explained in the introduction, if the points of $\mathbf{S}_X$ are Fréchet smooth regarded in $\mathbf{S}_{X^{\mathcal{U}}}$, then they are also that when considered as elements of the bidual. It is known that if the (bi)dual of a Banach space is Fréchet smooth, then the space is reflexive. We do not know what kind of geometric properties the Fréchet smoothness of $j(x)$, for all $x \in \mathbf{S}_X$ implies. For instance, in the case of M-embedded space X (see e.g. [9, 14]), if any $x \in \mathbf{S}_X \subset X^{**}$ is a Gâteaux smooth point, then it follows immediately that X is reflexive.

Theorem 2.6. *Let X be a superreflexive space. Then all Gâteaux smooth points $x \in \mathbf{S}_{X^{\mathcal{U}}}$ are in fact Fréchet smooth.*

Proof. According to the superreflexivity of X we have $(X^{\mathcal{U}})^* = (X^*)^{\mathcal{U}}$. Let $x \in \mathbf{S}_{X^{\mathcal{U}}}$ be a Gâteaux smooth point and let $f \in (X^{\mathcal{U}})^* = (X^*)^{\mathcal{U}}$, $\|f\| = 1$, be the unique exposing functional. Suppose that $(f_k) \subset \mathbf{S}_{(X^{\mathcal{U}})^*}$ is a sequence with $f_k(x) \rightarrow 1$.

By using the Smulyan lemma we will argue that x is a Fréchet smooth point, thus, assume to the contrary that $\limsup_{k \rightarrow \infty} \|f_k - f\| = C > 0$. By passing to a

Result. There is a reflexive, non-superreflexive Banach space X such that, for every free ultrafilter $\mathcal{U}$ on $\mathbb{N}$ and every $x \in S_X$, the canonical point $j_{\mathcal{U}}(x) \in S_{X^{\mathcal{U}}}$ is a Frechet smooth point of $X^{\mathcal{U}}$.

Idea of the Proof. Take an ℓ_2 -sum of finite-dimensional smooth spaces which become closer and closer to ℓ_{∞}^n . The summands force failure of superreflexivity, but the outer ℓ_2 -sum gives a strong orthogonality phenomenon in ultrapowers: after taking the weak ultralimit projection $\Pi : X^{\mathcal{U}} \rightarrow X$, the ultrapower splits isometrically as

$$X^{\mathcal{U}} = j_{\mathcal{U}}(X) \oplus_2 \ker \Pi.$$

Thus the canonical point $j_{\mathcal{U}}(x)$ is just $(x, 0)$ in an ℓ_2 -sum, and Frechet smoothness reduces to the ordinary Frechet smoothness of x in X .

Theorem 1. *Let $p_1 = 2$ and $p_n = \max\{2, \log n\}$ for $n \geq 2$, and put*

$$X = \left(\bigoplus_{n=1}^{\infty} \ell_{p_n}^n \right)_2.$$

Then X is reflexive and not superreflexive. Nevertheless, for every free ultrafilter $\mathcal{U}$ on $\mathbb{N}$ and every $x \in S_X$, $j_{\mathcal{U}}(x)$ is a Frechet smooth point of $X^{\mathcal{U}}$.

Proof. The space X is reflexive because it is an ℓ_2 -sum of finite-dimensional spaces.

It is not superreflexive. Indeed, for every n ,

$$\|a\|_{\infty} \leq \|a\|_{p_n} \leq n^{1/p_n} \|a\|_{\infty} \quad (a \in \mathbb{R}^n),$$

and $n^{1/p_n} \leq e$. Thus X contains subspaces uniformly isomorphic to ℓ_{∞}^n for all n . By the Enflo-Pisier finite representability criterion for superreflexivity, this rules out superreflexivity.

Each summand $\ell_{p_n}^n$, $1 < p_n < \infty$, is Frechet smooth away from zero. The ℓ_2 -sum norm on X is therefore Frechet smooth away from zero: for the squared norm this follows by differentiating the finite head and controlling the tail with

$$|\|a+b\|^2 - \|a\|^2| \leq 2\|a\|\|b\| + \|b\|^2$$

and Cauchy-Schwarz. Hence every $x \in S_X$ is a Frechet smooth point of X .

Fix a free ultrafilter $\mathcal{U}$. Since X is reflexive, every bounded sequence has a weak $\mathcal{U}$ -limit. Define the contractive projection

$$\Pi : X^{\mathcal{U}} \rightarrow X, \quad \Pi((x_m)^{\mathcal{U}}) = w\text{-}\lim_{\mathcal{U}} x_m.$$

Then $\Pi j_{\mathcal{U}} = \text{id}_X$. We claim that for every $x \in X$ and every $w = (w_m)^{\mathcal{U}} \in \ker \Pi$,

$$\|j_{\mathcal{U}}(x) + w\|^2 = \|x\|^2 + \|w\|^2. \quad (1)$$

Write $x = (x_r)_{r \geq 1}$, $w_m = (w_{m,r})_{r \geq 1}$, and let P_N be the projection onto the first N coordinates. Because each coordinate is finite-dimensional and $w \in \ker \Pi$, $P_N w_m \rightarrow 0$ in norm along $\mathcal{U}$. Therefore

$$\lim_{\mathcal{U}} \|P_N x + P_N w_m\|^2 = \|P_N x\|^2.$$

For the tail $Q_N = I - P_N$, if $\sup_m \|w_m\| \leq M$, then

$$|\|Q_N x + Q_N w_m\|^2 - \|Q_N x\|^2 - \|Q_N w_m\|^2| \leq 2M\|Q_N x\|.$$

Letting first $m \rightarrow \mathcal{U}$ and then $N \rightarrow \infty$ gives (1). Consequently

$$X^{\mathcal{U}} = j_{\mathcal{U}}(X) \oplus_2 \ker \Pi$$

isometrically.

Finally, a point $(x, 0)$ in an ℓ_2 -sum $X \oplus_2 Z$ is Frechet smooth whenever $x \in S_X$ is Frechet smooth. To see this, use the dual decomposition $X^* \oplus_2 Z^*$ and Smulyan's criterion: if $(f_k, g_k) \in S_{X^* \oplus_2 Z^*}$ satisfies $f_k(x) \rightarrow 1$, then $\|f_k\| \rightarrow 1$, $\|g_k\| \rightarrow 0$, and the normalized $f_k/\|f_k\|$ converge in norm to the unique support functional of x . Hence (f_k, g_k) converges in norm to the unique support functional of $(x, 0)$. Applying this to $X^{\mathcal{U}} = j_{\mathcal{U}}(X) \oplus_2 \ker \Pi$ proves that $j_{\mathcal{U}}(x)$ is Frechet smooth for every $x \in S_X$. $\square$

Limitations and novelty check. This does not settle whether the source hypothesis forces reflexivity or any other precise geometric property. It only eliminates superreflexivity as a necessary consequence. Cheap run indexes and web searches for the exact phrases “Frechet smooth canonical ultrapower”, “Frechet smooth ultrapower superreflexive”, and variants did not reveal this explicit example as an existing answer.

References

- [1] J. Talponen, *Uniform-to-proper duality of geometric properties of Banach spaces and their ultrapowers*, arXiv:1412.1279; Math. Scand. **121** (2017), 111–120.
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